

★ LEVEL 1 — BASIC / THEORY TYPE (120 QUESTIONS)

(Covers ALL topics mixed: Relations, Types of Relations, Functions, Types of Functions, Domain–Range, Composition, Invertible Functions)

A. VERY SHORT ANSWER (1–2 lines)

(1–40)

1. Define a relation between two sets.
2. What is the domain of a relation?
3. What is the range of a relation?
4. What is codomain?
5. Define reflexive relation.
6. Define symmetric relation.
7. Define transitive relation.
8. What is an equivalence relation?
9. What is a partial order relation?
10. Define identity relation with an example.
11. Define universal relation with an example.
12. Define antisymmetric relation.
13. What is a function?
14. Is every relation a function? Why?
15. Define domain of a function.
16. Define range of a function.
17. Define codomain of a function with example.
18. Define one-one function.
19. Define onto function.
20. Define one-one onto function.
21. Define many-one function.
22. Define into function.
23. Define constant function.
24. Define identity function.
25. Define polynomial function.
26. Define modulus function.
27. Define signum function.
28. What is greatest integer function?
29. What is least integer function?
30. Define logarithmic function.
31. Define exponential function.
32. Define invertible function.
33. Define inverse of a function.

34. What is composition of functions?
 35. When is a function invertible?
 36. What is mapping diagram of a function?
 37. What is the vertical line test?
 38. What is the horizontal line test?
 39. What is composite relation?
 40. Are all equivalence relations symmetric?
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B. SHORT ANSWER (3–4 lines)

(41–80)

41. Write an example of a relation which is reflexive but not symmetric.
42. Write an example of a relation which is symmetric but not transitive.
43. Write an example of a relation which is transitive but not reflexive.
44. Show that the universal relation is reflexive.
45. Give an example of an equivalence relation from real life.
46. Explain antisymmetric relation with an example.
47. How many relations exist from $A=\{1,2\}$ to $B=\{3,4,5\}$?
48. Explain why identity function is one-one and onto on a set.
49. Give an example of a many-one into function.
50. Give an example of an onto function but not one-one.
51. Explain invertible function with an example.
52. What is graphical representation of a function?
53. Why is $\sqrt{x}$ not defined for negative x ?
54. What is the range of $y = |x|$?
55. State domain and range of $y = 1/x$.
56. State domain and range of $y = \sqrt{4-x^2}$.
57. Is $y = x^2$ one-one on $\mathbb{R}$? Explain.
58. Is $y = x^3$ invertible? Why?
59. Explain composition of functions with example.
60. Show that composition of two functions may not be commutative.
61. What is the domain of $f(x) = \log(x-2)$?
62. What is the domain of $f(x) = \sqrt{x+5}$?
63. Find the domain of $f(x)=1/(x^2-9)$.
64. If $f(x)=|x|$, find $f(-3)$.
65. If f is one-one and g is one-one, is $g \circ f$ one-one?
66. Describe the relation $R=\{(1,2),(2,1),(1,1),(2,2)\}$ using a digraph.
67. How many equivalence relations can exist on a 2-element set?
68. Explain partition corresponding to equivalence relations.
69. Explain parity relation on integers.

70. What is the domain of $f(x)=\tan^{-1}(x)$?
 71. If $f:A\rightarrow B$ is onto, what can be said about range and codomain?
 72. Give mapping of MANY-ONE function example.
 73. Explain how a graph can show one-one function.
 74. Is every function from $\mathbb{R}$ to $\mathbb{R}$ invertible?
 75. Explain why $f(x)=\sin x$ is not invertible.
 76. How can we make trigonometric functions invertible?
 77. Find domain of $f(x) = \sqrt{5-3x}$.
 78. Find range of $f(x) = x^2 + 4$.
 79. Does $f(x) = |x|$ have inverse? Explain.
 80. What is algebra of functions?
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C. LONG ANSWER (5–8 lines)

(81–120)

81. Describe reflexive, symmetric, transitive relations with examples.
82. Show that "is equal to" relation is equivalence.
83. Show that "divides" relation on $\mathbb{N}$ is antisymmetric and transitive.
84. Show that the relation $R = \{(a,b): a-b \text{ is even}\}$ is equivalence.
85. What is the difference between equivalence and partial order?
86. Write all possible relations on $A=\{1,2\}$.
87. Prove that a function can never assign two images to the same element.
88. Show that $f(x)=1/x$ is neither even nor odd.
89. Explain even and odd functions with graph.
90. Show that a one-one onto function has a unique inverse.
91. Explain composition of functions with three different examples.
92. Show that composite of invertible functions is invertible.
93. Explain why $\sqrt{x^2}=|x|$.
94. Determine if $f(x)=x^3-1$ is invertible.
95. Show that $f(x)=x^2$ is not one-one but becomes one-one on $x\geq 0$.
96. Explain onto functions using mapping diagrams.
97. Find domain and range of $f(x)=1/\sqrt{4-x}$.
98. Describe the significance of codomain vs range.
99. Explain number of possible functions from A to B .
100. Write all functions from $A=\{1,2\}$ to $B=\{3,4\}$.
101. Show that composition of two constant functions is constant.
102. Give example of composition that is defined but reversed composition is not.
103. Prove that if f is invertible, then $f\circ f^{-1} = \text{identity}$.
104. Prove that identity function is invertible.
105. Find domain and range of $f(x)=\sqrt{x^2 + 1}$.

106. Show that $f(x)=|x|$ is not invertible.
107. Show that $f(x)=x^3$ is invertible and find inverse.
108. Show that inverse of one-one onto is unique.
109. Explain division of real numbers and its domain restrictions.
110. Show that $f(x) = \sin^{-1}x$ has domain $[-1,1]$.
111. Explain continuity of basic functions (conceptual).
112. Show that $f(x)=\tan x$ is not defined at odd multiples of $\pi/2$.
113. Find domain of $f(x)=\sqrt{2x-1}+1/(x^2-4)$.
114. Show that $f(x)=\log x$ is invertible and find inverse.
115. Prove that constant function cannot be invertible.
116. State 5 examples of functions used in physics.
117. Compare injective vs surjective with diagrams.
118. Show that relation $R=\{(1,1),(2,2),(3,3)\}$ is equivalence.
119. Explain relation as a subset of Cartesian product.
120. Explain why relation may not be function but function is always relation.